



Fig. 2: Pin details of regulator 7805 and transistor BC548

microcontroller. Port pins RA0, RA1 and RA2 receive inputs to monitor battery voltage, charge current and solar panel voltage, respectively, to control the overall process and display information on the LCD module. When port pin RA3 goes high, transistor T1 becomes saturated and relay RL1 energises to connect the solar panel to the battery.

Regulator 7805 provides regulated 5V to the microcontroller and the LCD module. Fig. 2 shows the pin details of regulator 7805 and transistor BC548.

This solar charger can charge the battery in two modes—boost and trickle. If battery voltage is greater than 12V the battery is charged in trickle mode, whereas if battery voltage is less than 12V it is charged in boost mode. In trickle mode, the battery is charged at discharge rate.

The system also calculates the energy that has been received from the solar panel. This gives an indication of the power that can be harnessed from the Sun.

Construction and testing

An actual-size, single-side PCB for the PIC16F877A microcontroller-based solar charger is shown in Fig. 3 and its

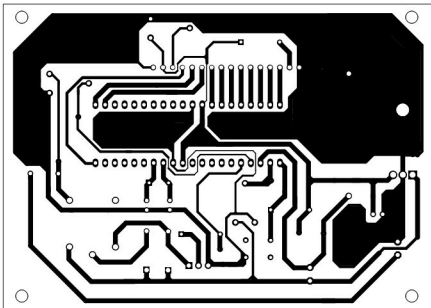


Fig. 3: An actual-size, single-side PCB for PIC16F877A microcontroller-based solar charger

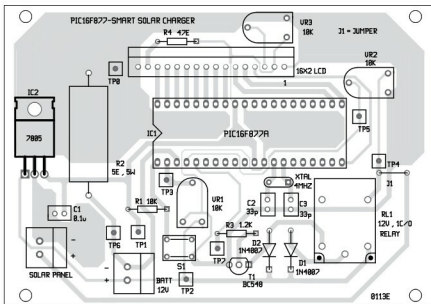


Fig. 4: Component layout for the PCB

PARTS LIST	
Semiconductors:	
IC1	- PIC16F877A microcontroller
IC2	- 7805, 5V regulator
T1	- BC548 npn transistor
D1, D2	- 1N4007 rectifier diode
LCD	- 16-character x 2-line LCD module
Resistors (all 1/4-watt, 45% carbon):	
R1	- 10-kilo-ohm
R2	- 5-ohm, 5W
R3	- 1.2-kilo-ohm
R4	- 47-ohm
VR1-VR3	- 10-kilo-ohm preset
Capacitors:	
C1	- 0.1μF ceramic disk
C2, C3	- 33pF ceramic disk
Miscellaneous:	
S1	- Push-to-on tactile switch
RL1	- 12V, single-changeover relay
X _{AL}	- 4MHz crystal
BATT.	- 12V lead-acid battery
	- Solar panel



Fig. 5: Author's prototype

component layout in Fig. 4. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base for microcontroller PIC16F877A. Before inserting the IC, check the supply voltage (5V) at test point TP1.

Before using the circuit, the system has to be calibrated for battery and solar voltages. This is done as follows: